TennisSetMaker – Version 1.0 Scope

Document Info

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1. Project Overview

1.1 Project Background

Warren Vickers, Director and Head Coach at TennisHQ, identified operational issues with existing set-making software unable to generate good variety of balanced doubles set across sessions in a way that reflects how organisers think about fairness and social mixing. Organisers currently rely on tools that don't account for constraints that matter to them specifically.

1.2 Project Objective

Develop a web-based application tool that generates fair, balanced, and socially engaging doubles pairings for tennis doubles, with manual configurable control and settings for users.

The features of **Version 1.0** enable club administrators to:

- Automatically generate a full session's worth of doubles sets, balancing skill and social variety
- Configure session parameters: number of courts, rounds, session duration, and start time
- Assign same-gender or mixed-gender sets, with automatic relaxation when a strict format would leave players unplaced
- Ensure fair rotation of sit-outs when player numbers are odd or unbalanced
- Manually override any automatically generated set, with overrides taking priority over the algorithm
- Manage player records (create, edit, remove; duplicate prevention)
- Produce easily printable schedules and QR codes for players to view court assignments

2. Stakeholders

- Warren Vickers (TennisHQ) – Client representative
- TennisHQ Staff and Players – End Users
- UWA Cits3200 student team 5 – Developers
- UWA unit coordinator/staff – Academic supervisors

3. Core Features

3.1 Set Matching Algorithm System

The core system of the application, automatically generates doubles sets for entire session, following presets of priority orders that balances fairness, social diversity, and competitive skill.

Priority Design

Build as independent, reorderable “blocks” rather than fixed sequential logic, each preset is at it’s core an ordered, toggleable list of which “blocks” apply. Letting multiple users/clubs to enable/disable/reorder priorities without changing the algorithm’s code in the future.

Note: Version 1.0 of the application will not support creating custom algorithm presets as extensive testing is required for accurate results.

Priority Logic

Hard requirement:

- The player has signed up/registered for the set
- The court is available and usable at the scheduled time
- Any court preference marked mandatory (rather than soft) is respected

Priority Weights: (default preset)

1. Sit out ratio
2. Set format compliance (mixed/non-mixed)
3. Social variety – unique partners
4. Social variety – unique opponents
5. Skill balance
6. Court preferences

7. Requested number of sets (available, disabled by default – see note below)

Note: Priority 7 is available for clubs weighting fairness by requested playtime rather than raw sit-out ratio. Off in TennisHQ's default preset per client direction, sit-out ratio is treated as fair regardless of how many sets a player signed up for.

Requirements

- Algorithm automatically generates a full round of doubles set given the current player pool and session configuration
- Algorithm respects priority hierarchy preset
- Algorithm regenerates sets correctly each round, accounting for new saved variables (partners, opponents, sit-out ratio)

Success Criteria

- Generates accurate and correct set pairings
- All hard requirements are accurately respected

3.2 Session & Court Configuration System

Administrators can configure the structure of session before sets are generated based on pre-configured and user-configured presets to streamline functionality and efficiency.

Requirements

- Admins can configure or use presets for number of active courts, number of rounds, session duration, and start time
- Admins can select set format for each round (mixed/non-mixed), with automatic relaxation if a strict format would leave players unplaced

Success Criteria

- Configured session correctly generates the specified number of rounds and sets

3.3 Manual Control System

Administration system to adjust and edit the generated sets manually and override any pairing results generated by the algorithm. Manually overridden sets are treated as final for that round and are not altered by the algorithm. The outcome of an override (updated partner history, sit-out counts, etc) is carried forward and factored into the next round of generation.

Requirements

- Admins can edit any automatically generated set at any point
- Manual overrides are considered by subsequent algorithm runs for that round

Success Criteria

- A manual change to a set persists and is reflected in the published printed/QR schedule
- Algorithm does not revert admin override

3.4 Result Output & QR Access

Generated sets should be easily publishable for distribution to players with option to store in a database for administration purposes.

Dependencies

- QR Code generator **(see 7.1)**

Requirements

- The system generates a printable schedule of all sets/court assignments for a session
- The system generates a QR code players can scan to view their court assignment

Success Criteria

- Generated schedule accurately reflects final (post-override) set assignments
- QR code correctly links to, or displays, the relevant court assignment information

3.5 User Account System

Users created account system to manage and edit player/court database, algorithm presets, restrict unauthorised access.

Users Must Be Able To

- Register an account
- Log in using email/username and password
- View account setting page
- Update account data (username, password, email)
- Update club data (remove, add, edit) **(see 3.6)**
- Log out of their account
- If opted in maintain logged in across sessions

Data Requirements

- Email
- Username
- Password
- UniqueID (For shared database)

Success Criteria

- Users can successfully register and login
- Passwords are securely stored using hashing/salting
- Invalid login attempts return appropriate error messages
- User sessions persist if user opts to stay logged in

3.6 Club/Organisation management

Overview

Ownership is scoped per-club and not per-user account, a single user can simultaneously be

the owner of their own created club, whilst holding roles in one or more other clubs they've been invited to. A user can also create and own multiple clubs of their own and invite other users as Admin or Staff.

Invitations

Club owner can add people in two ways:

- Invite an existing user – if the person already has an account on the application platform, the Owner invites them by email or username, and the invite appears for them to accept or decline.
- Invite via link – if the person does not have an account on the application platform, the Owner invites them by email through the same process, and the invitation generates a signup link. On registration through that link, the new user is automatically added to the inviter's Club under the rank the Owner selected when creating the link.

Data Requirements

- Club name
- Club users (all roles)
- User roles (owner, admin, staff)

Requirements

- Owners can create, edit, and remove clubs from their account
- Owners can edit club configurations, data and presets
- Owners can add/remove other existing/non-existing users to their club
- Owners can assign/remove admin or staff role to club users
- Admins can edit club configurations, data and presets
- Admins can add other existing/non-existing users to the club
- Admins can assign other users as admin/staff but cannot revoke admin role.
- Staff can use club configurations, presets and view data

Success Criteria

- Owners can create, edit, and remove a club without error
- Club configuration, data, and preset changes made by Owners/Admins persist correctly
- Invited users (existing or new) receive and can accept/decline an invite correctly
- New users registering via an invite link are automatically added to the correct club under the assigned rank
- Role assignments (Owner/Admin/Staff) are correctly enforced across the platform
- Admins are blocked from revoking another Admin's role
- Staff cannot edit club configurations, data, or presets
- Removed users immediately lose access to the club's data

3.7 Player/Court Management System

Administration tools to edit and maintain the club's player roster, court listing, club listings and view player sit-out fairness statistics.

Data Requirements

- Player name
- Player gender
- Player skill level
- Player Sit-out ratio/history
- Court number
- Court type
- Court status (active/inactive)

Sit-out frequency/ratio

Should be dynamically tracked based on activity over timeframe to accurately accommodate player.

Requirements

- Admins can create, edit, and remove player records
- Admins can create, edit, and remove active courts
- System prevents duplicate player records where possible

Success Criteria

- Admins can alter records without error
- Duplicate player entries flagged
- Records persist correctly between sessions
- Admins can view individual player sit out frequency

4. User Experience

The platform should prioritise simplicity, functionality and speedy design.

4.1 Expected Typical Use

1. User opens platform
2. User clicks “Set generator”
3. User reviews/adjust presets configurations (players, courts, time, number of sets)
4. User clicks “Generate”
5. User reviews/adjust results
6. User clicks “Post”
7. Results adjust/stores respective data and “Print” prompt appear

4.2 Design Goals

- Easy to use no over the top multi layered interfaces
- Clear, simple and functional interface
- Mobile accessible layout for QR scan results
- Admins should be able to identify lowest sit-out frequency players easily
- Fast process

5. Technical Components

5.1 Frontend

User facing systems and interfaces for interaction with the platform and application, should be aesthetically clear, simple to use and functionally fast and responsive.

Respective Pages/Interfaces

- Homepage interface
- Login interface
- Signup interface
- Account setting interface
- Set generator interface
- Set result preview interface
- Set result interface
- Preset configurations interface
- Player/Court management interface
- Player/Court Statistics interface (sit-out ratio)

Tools And Technologies

- Python
- HTML
- JavaScript
- CSS
- Tailwind/Bootstrap

5.2 Backend

Behind the scenes infrastructure and system operating the platform and application, should be well designed and documented for future proofing and expansion.

Tools And Technologies

- Python
- C
- Flask
- Jinja

5.3 Database

Integrate with existing databases for data migration and management cleanly. Accommodate role-based access control on shared databases to allow multiple users under the same organisation to access and manage relevant data.

Data Stored

- Store user account data (email, username, password)
- Club/organisation role records (owner, admin, staff)
- Player data (name, gender, skill level, sit-out ratio)

- Court data (court number, type, active/inactive status)
- Configuration presets (session time, courts, format)

Tools And Technologies

- PostgreSQL

6. Non-functional Requirements

6.1 Security

The system must:

- Hash/Salt user passwords
- Have secure data sanitisation
- Authenticate users and restrict access to protected pages
- Randomise set/session pairing result (QRcode URL token)

6.2 Reliability

The system should:

- Support multiple browsers (chrome, firefox, safari, etc.)
- Support multiple devices (desktop, phone, tablets)
- Handle system failures gracefully
- Display clear error messages when issues occur

7. Key Library Dependencies

7.1 QR Code Generator (Lib)

Generated locally on the server at request/print time, no external service dependency, created in memory from a session/results URL and embedded directly into the printable schedule.

- qrcode (Python library)

8. External Services

8.1 No external services

The application does not currently plan to use any external services.

9. Out of Scope features

The following features are excluded from version 1.0 scope:

- Offline build [low priority]
- User configurable algorithm presets (flexible foundation) [low priority]
- Expandable to other sports (flexible foundation) [low priority]
- User preferences submission [low priority]

- Automated notifications (admins, players) [low priority]

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